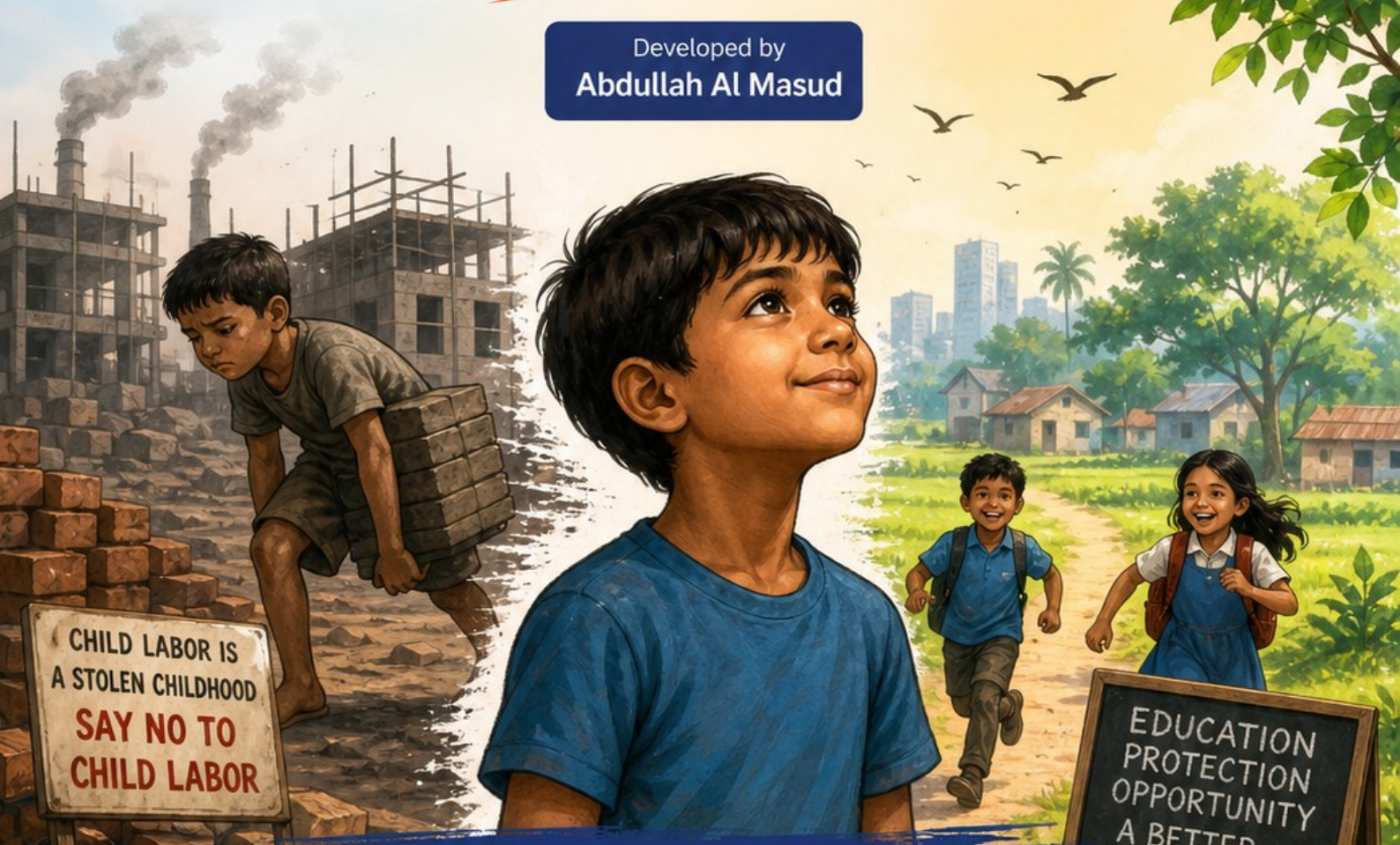
মুক্তপাখি

Mkto Pakhi

Protect child from forced labor and abuse

Developed by
Abdullah Al Masud



Every Child Deserves a Childhood,
Not Child Labor

Supported organizations

UNDP Bangladesh

ICT Division

BCC

Embassy of Norway in Bangladesh

CHAPTER I

Introduction

1.1 Introduction

Child labour continues to be a serious and widespread problem in Bangladesh, deeply affecting the lives and future of countless children across the country. According to the latest 2025 survey conducted jointly by the Bangladesh Bureau of Statistics (BBS) and UNICEF, about 9.2% of children aged 5–17 years are engaged in child labor[1]. A noticeable rise from 6.8% in 2019 which means an additional 1.2 million children are now involved[2]. Poverty, weak enforcement of existing labor laws, and the long-term effects of the COVID-19 pandemic have forced many families to depend on their children's earnings. As a result, large numbers of children are working in dangerous sectors such as agriculture, garment factories, and domestic households, facing serious risks to their health, safety, and overall development while being denied the right to education. In response to this pressing challenge, the project titled "Mukto Pakhi - A Child Safety Android Application" has been developed as a practical and innovative mobile tool. The main objective of Mukto Pakhi is to create a safer environment for children by enabling quick reporting of child labor cases, providing location-based tracking, and raising awareness among parents, communities, and local authorities through simple and accessible features. The application has been built using the Flutter framework, which allows smooth and efficient development for Android devices. Mukto Pakhi includes several important features supported by reliable libraries and plugins, such as secure user login and real-time data storage using Firebase Authentication and Cloud Firestore, location tracking and alerts through the geolocator plugin, instant emergency notifications plugin, photo capture for evidence using the camera plugin, and smooth user experience managed by the provider package.

1.2 Motivation

The motivation behind developing the Mukto Pakhi Application stems from a strong desire to tackle one of Bangladesh's most pressing social issues: child labor. During my internship experience, I witnessed how technology can be a powerful tool for solving real-world problems, particularly in a country like Bangladesh where child labor affects millions of families due to poverty and limited access to education. This inspired me to choose Flutter as the development framework because it allows for quick and efficient building of mobile apps that work well on Android devices, which are widely used globally nowadays. Flutter's single codebase approach saves time and resources, making it ideal for creating user-friendly applications that can reach more people without high costs. By focusing on this app, I aimed to provide practical features like real-time reporting, location tracking, and awareness modules to help parents, communities, and authorities protect children from exploitation. Ultimately, Mukto Pakhi represents my commitment to using technology for positive change, contributing to national efforts to reduce child labor and build a better future for Bangladesh's young generation.

1.3 Objective

The primary objective of developing the Mukto Pakhi Application is to create an effective, user-friendly mobile tool that helps protect children in Bangladesh from forced labour, abuse, and exploitation. One of the key ways Mukto Pakhi helps rescue children from forced labour and abuse is by providing quick and reliable tools for detection and response. This not only protects the child from immediate harm but also connects them to support services like local NGOs for counseling, education, and family help, breaking the cycle of abuse. This makes the app easy to update and maintain, ensuring it reaches more users without needing advanced skills or expensive resources.

The key objectives of this project can be summarized as follows:

1. To enable real-time reporting of child labour and abuse with photo, legal documents and location evidence.
2. To connect all volunteers in a community group and working on reports by location and region.
3. To take legal action by administrative officials or NGO's people through analysis and verification of each reported case, and later ensure the rehabilitation, food and education of the children.
4. The data of each member of the community will be protected in our database and can be retrieved or updated for later as user's need.

CHAPTER II

Methodology

The motivation for developing Mukto Pakhi as a Flutter-based Android application comes from the need to create an effective and accessible tool that can truly help solve the widespread problem of child labour in Bangladesh. During my internship, I learned that technology, especially mobile apps, can make a real difference in everyday challenges like protecting children from exploitation. I chose Flutter because it is a modern framework that lets developers build high-quality apps quickly and at low cost, using a single codebase that works smoothly on Android phones which are affordable and commonly used by people in rural and urban areas of Bangladesh. This makes the app easy to update and maintain, ensuring it reaches more users without needing advanced skills or expensive resources. Flutter's features, such as fast performance and beautiful interfaces, help make the app user-friendly for parents, community members, and authorities who may not be very tech-savvy. By focusing on this technology, Mukto Pakhi becomes a practical solution that empowers ordinary people to act against child labour, turning motivation into real action for a safer society.

One of the key ways Mukto Pakhi helps rescue children from forced labour and abuse is by providing quick and reliable tools for detection and response. For example, if a child is spotted working in harmful conditions like factories or fields, users can use the app's camera feature to take photos as evidence and report it instantly from their location. The geolocator plugin tracks the exact spot, which helps rescuers find the child faster and remove them from danger.

Instant reporting in Mukto Pakhi works through a connected community system that brings everyone together for better protection. When a user sees a case of child labour, they can fill in simple details in the app and submit the report, which goes to a shared database using Cloud Firestore. This transparent system builds trust, encourages more reporting, and helps enforce laws more effectively, leading to penalties for violators and long-term prevention of child labour.

2.1 Project Flow Chart

The project flowchart for Mukto Pakhi – A Child Safety Android Application provides a clear overview of how users navigate through the app, starting from the beginning and covering all main features in a logical sequence. This helps in understanding how each module or feature relates to others and how users navigate within the application.

Fig. 2.1: Project Flow Chart

It begins at the "Start" point, where the app checks if the user already has an account. If not, the flow directs them to the "Register" page for creating a new profile with basic details like name, phone number, and password, ensuring secure access via Firebase Authentication. If the user has an account, they proceed to "Community Membership Access," which verifies their login and grants entry to the core functions, promoting a sense of community involvement right

from the start. From there, the flow leads to the "Home Page," which acts as the central hub, branching out to various features like evidence collection, emergency calls, community contacts, maps, feedback, and settings.

One key branch from the Home Page is the "Direct Report" decision point, which allows users to quickly decide if they want to file a report immediately. If "Yes," the flow goes to "Share Location," where the geolocator plugin captures the user's current position for accurate incident mapping. Then, users can "Select Member" from their community contacts to involve trusted people, and finally submit the "Report." If "No" to direct report, the flow shifts to the "Report Page," where users "Choose an Action" such as adding details about the child labour case, share evidence from gellary like picture or allow direct phone camera and then proceed to "Report." This feature ensures fast reporting of child abuse or forced labour, with options to attach photos using the camera plugin or documents from file manager to helping authorities respond quickly with evidence.

Other important features accessible from the Home Page include "Evidence," which lets users gather and store proof like images or notes for later reports, making sure nothing is missed during emergencies. The "Call 109" button provides a direct link to Bangladesh's national child helpline for immediate voice support, integrating with the phone's dialer for urgent cases. "Community Contact" connects users to a list of local volunteers, NGOs, or guardians, allowing quick messaging or calls through the app's integration, building a supportive network. The "Map of Reported Case" shows a visual overview of ongoing or past reports on a map, using location data to highlight hotspots of child labour, which helps communities stay alert and authorities plan actions.

Additional tools from the Home Page are "Users Feedback," where people can share suggestions or rate the app to improve it over time, and "Settings," which branches into options like "Log out" for secure exit, "Set Guardian" to directly add community member by using email, phone and address, and "Update Profile" to change personal details. These ensure the app remains user-friendly and customizable. Overall, the flowchart highlights how Mukto Pakhi combines technology with community effort to protect children, from initial access to detailed actions, making it a reliable tool for real-world impact in Bangladesh.

2.2 Tools and Services

The app follows a clear and simple flow, as shown in the flowchart, starting from the home screen where major features are designed to aware user about how action quick and eye-catching design and child labor representation pictures to get users emotions to appreciate our app design motivation to stay connected with us for protecting children and guiding user step by step to use its powerful features. Every service and tool is carefully designed to be easy, fast, and effective, even for people with basic smartphones and limited internet.

2.2.1 Firebase Firestore

The user registration and login process in the Application is designed to be very simple and user-friendly so that people from different backgrounds in Bangladesh can easily join and use the app to protect children. When a new user opens the app for the first time, they see the "Create Profile" screen. Here, they need to fill in five basic details: Name, Phone number, Email, Password, and Address.

For users who already have an account, the process is even easier. They go to the "Login" screen, enter only their Email and Password, and press the "Login" button. The app quickly checks these details against the stored data in Firebase Firestore.

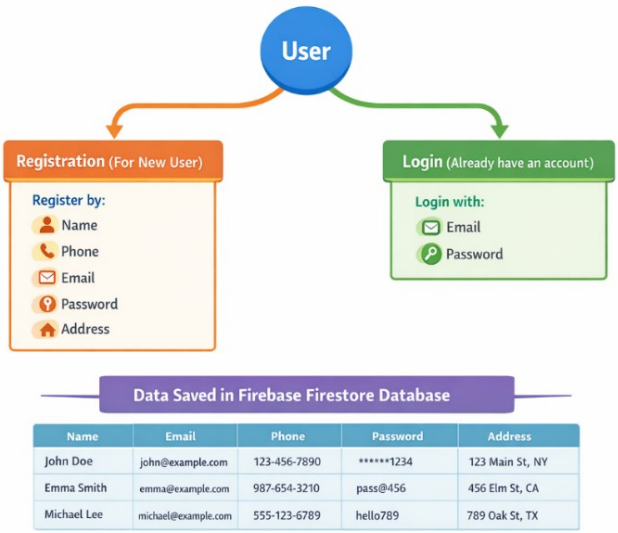


Fig. 2.2: Firebase Firestore

All user data is stored in Firebase Firestore in a very organized way. Each registered user gets their own document inside the "users" collection, and this document has exactly five main fields:

1. Name
2. Phone (with country code)
3. Email
4. Password (stored securely in encrypted form)
5. Address

Firebase Firestore is highly secure because it uses strong encryption, requires authentication for every access, and follows Google's strict security rules. This means no one can see or change another user's personal information without proper permission. At the same time, the app allows registered users to update their details anytime. From the Profile section in Settings, users can edit their phone number, address, or other information and save the changes. The updated data is immediately sent back to Firestore, keeping the user's profile fresh and accurate. This flexibility makes Mukto Pakhi safe, reliable, and practical for everyday use in our country.

2.2.2 Plugins & Packages

Mukto Pakhi was developed using the Flutter framework along with several reliable plugins to ensure smooth, fast, and trustworthy performance on Android devices commonly used in Social and Safety application. Key plugins include:

a. image_picker

The **image_picker** plugin is a reliable and widely-used Flutter package that enables users to easily select or capture images directly within the Mukto Pakhi app. It is implemented in two main files: **child_labor_service.dart** (for attaching evidence during incident reporting) and **evidence.dart** (for standalone evidence collection).

In Mukto Pakhi, the camera plugin(image_picker) is mainly used in the reporting section (child_labor_service.dart) and evidence collection page (evidence.dart). When a user wants to report child labour or abuse, they can press the "Take Photo" button, which opens the phone's camera instantly. After taking the photo or video, it is automatically attached to the report and saved securely in Firebase Storage for authorities to review.

Using the simple method **ImagePicker().pickImage()**, the plugin allows users to either:

- Take a new photo using the phone's camera, or
- Choose an existing photo from the gallery.

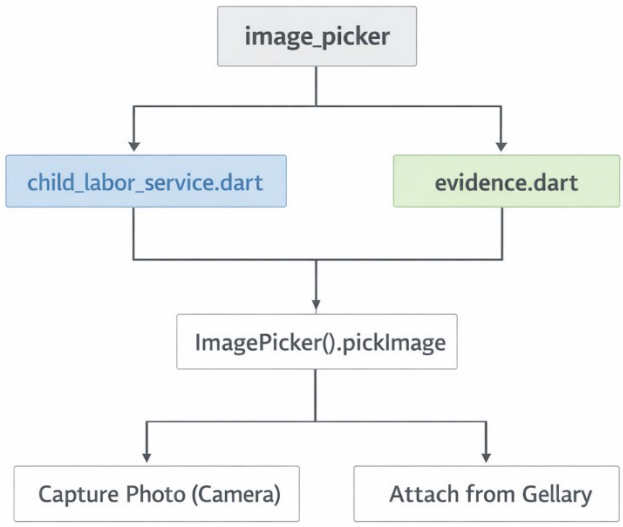


Fig. 2.3: image_picker plugin

This feature is very important because it lets users quickly attach clear visual proof (photos of child labour situations, working children, or abuse) when submitting reports. By making evidence collection fast and user-friendly, **image_picker** helps strengthen reports so authorities and NGOs can act more effectively and rescue children faster.

b. flutter_osm_plugin:

The flutter_osm_plugin is a powerful and open-source Flutter package used in Mukto Pakhi to display interactive maps without depending on expensive or restricted services like Google Maps. This plugin is implemented in the map.dart file and is the backbone of the "Map of Reported Cases" feature.

How it works in Mukto Pakhi:

The plugin uses OSMFlutter widget to show real-time OpenStreetMap (a free, community-built map similar to Google Maps). A MapController object controls the map's behavior. When the user opens the map screen:

- The app automatically gets the user's current location using the geolocator plugin and centers the map there.
- It then adds markers (small pins) on the map to show locations of all previously reported child labour or abuse cases, pulled from Firebase Firestore in real time.
- Users can zoom in/out, drag to move the map, and tap markers to see details like report time, description, or status.

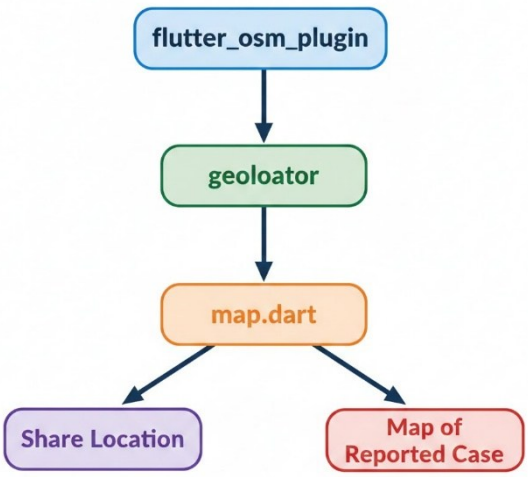


Fig. 2.4: flutter_osm_plugin

Why this plugin was chosen for Mukto Pakhi:

- No API keys or costs – Unlike Google Maps, OpenStreetMap is completely free and does not require any billing setup or API restrictions, making it perfect for a low-budget, community-focused app in Bangladesh.
- Offline capability – The map tiles can be cached, so it still works reasonably well in areas with poor or no internet.
- Full control – The plugin allows easy adding, moving, and customizing of markers, which is essential for showing child labour hotspots clearly.
- Lightweight and fast – It performs smoothly even on low-end Android phones commonly used in rural and urban Bangladesh.

By using flutter_osm_plugin, Mukto Pakhi provides an easy-to-understand visual overview of reported cases, helping communities stay alert, authorities plan rescues, and users feel more connected to the child protection effort.

c. url_launcher

The url_launcher plugin is a lightweight, secure, and highly trusted Flutter package that allows Mukto Pakhi to safely interact with external applications already installed on the user's Android phone. It is used in four important files:

- call.dart: To open the phone dialer for emergency calls (e.g., to the national helpline 109 or a trusted volunteer).
- contact_volunteers.dart: To directly dial or message community volunteers and local NGO contacts for immediate support.
- child_labor_service.dart: To launch the email client with pre-filled details (recipient, subject, report summary) so users can send formal complaints or evidence to authorities.
- share_location.dart: To open the default messaging app with pre-filled location link and message, making it easy to share the incident spot with family, guardians, or helpers.

This plugin works by using special URI schemes (such as tel:, sms:, and mailto:) to tell the phone which app to open and what information to pre-fill. For example, when a user taps “Call 109”, the plugin simply passes tel:109 to the system, and the native dialer opens instantly with the number ready. Because it hands off the action to the phone’s built-in apps instead of handling calls or emails inside the app, it remains very safe, respects user privacy, and feels natural to everyone using this app.

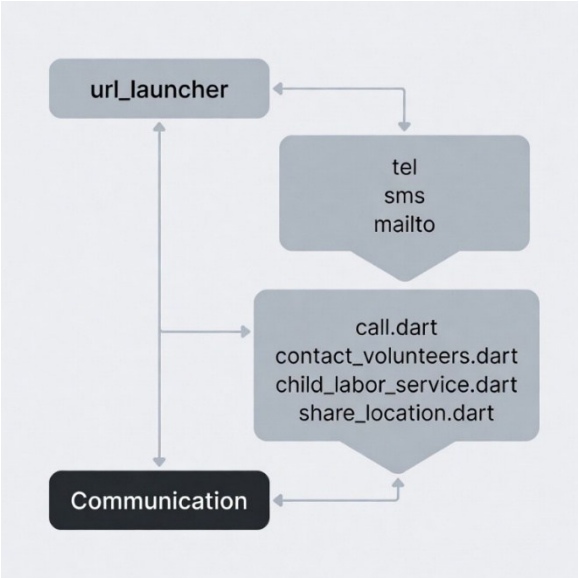


Fig. 2.5: url_launcher plugin

How url_launcher works in Mukto Pakhi: This plugin uses special link formats called URIs to connect the app to the phone’s built-in tools. The main uses are:

- tel: URI – opens the phone dialer and automatically fills in a number (e.g., 109 for the national child helpline or a volunteer’s phone number) so the user can call for urgent help without typing manually.
- sms: URI – opens the messaging app with a pre-filled phone number and sometimes a ready message, making it faster to contact community members or NGOs.
- mailto: URI – opens the default email app with pre-filled recipient address, subject, and body text (e.g., a ready report summary with evidence links), helping users send detailed information to authorities or organizations.

So, url_launcher makes emergency calls, volunteer contacts, and official communications quick and effortless. By letting users reach help through their phone's familiar tools in just one tap, it plays a vital role in speeding up child rescue and protection efforts.

d. permission_handler_android

The permission_handler package (from pub.dev) is the standard cross-platform solution for requesting, checking, and managing runtime permissions in Flutter apps. It is used to request and manage runtime permissions on Android devices.

- permission_handler_android is the Android-specific implementation inside this package (you don't import it directly; the main permission_handler package handles platform differences).
- On Android, many features require explicit user permission at runtime (not just in the manifest).

Common permissions it manages on Android include:

- Camera (for taking photos of cases)
- Storage / Photos / Media (for uploading evidence images)
- Location (for live location sharing in reports)
- Phone (for making direct calls to authorities)

So, the permission_handler_android is the Android-specific implementation package of the permission_handler plugin. That is used to request and manage runtime permissions on Android devices.

e. shared_preferences.dart

The shared_preferences.dart is used for local data storage in a Flutter app. It allows the app to store small amounts of data on the user's device persistently (data remains even after the app is closed). shared_preferences is a very popular Flutter plugin (from pub.dev) that provides a simple way to store and read small amounts of persistent key-value data on the device.

Characteristics of the important plugin in Flutter:

- User login status (e.g., "isLoggedIn": true)
- App preferences (report edit, membership update)
- Small settings (last viewed reported case, notification enabled)
- Simple counters or flags

In Mukto Pakhi, shared_preferences is perfect for remembering if a volunteer has seen an onboarding tutorial, storing a user token for authentication, saving preferred notification settings, or keeping track of the last reported case timestamp all without needing a full backend or database.

f. material.dart

The material.dart provides Material Design UI components for Flutter. It is the core UI library used to build Android-style Flutter apps. It provides all the Material Design widgets and components that follow Google's Material Design guidelines (now Material 3 by default in recent Flutter versions).

Key elements include:

- Widgets like Scaffold, AppBar, Button (ElevatedButton, TextButton, etc.), Card, ListTile, TextField, Dialog, BottomNavigationBar, FloatingActionButton, Theme, and many more.
- Themes, colors, typography, elevations, shapes, ink ripples, animations, and accessibility features.
- Scaffold: The fundamental layout structure with AppBar, body, FloatingActionButton, Drawer, BottomNavigationBar/NavigationBar, etc.
- MaterialApp: The root widget for most apps, sets up routing, theming, localization, and Material-specific navigation.
- Input & Forms: TextField and DropdownButton, Switch, Checkbox, Radio, Slider

Almost every modern Flutter app starts with this import because it gives you beautiful, consistent, Android-native-feeling UI out of the box.

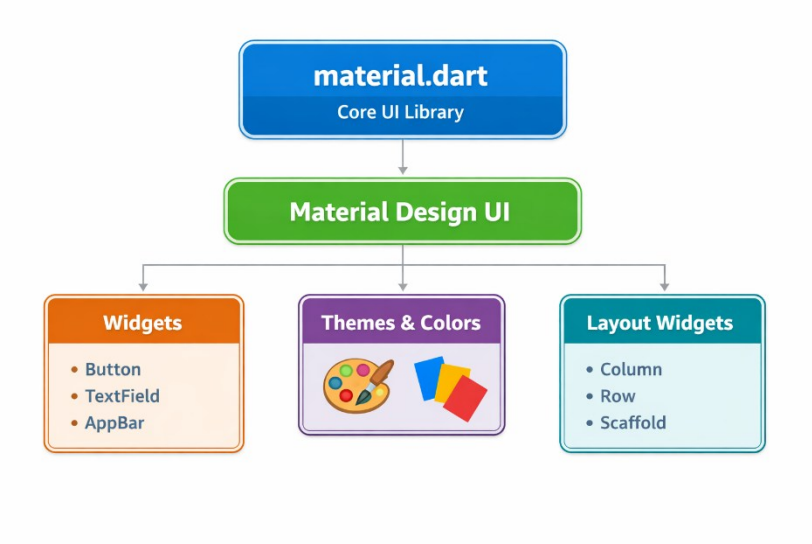


Fig. 2.6: Flutter UI material package

In Mukto Pakhi, you use material.dart to build clean screens for community groups, report forms, maps, photo previews, and alert dialogs ensuring the app looks professional and follows Android design standards.

2.3 Flutter Framework

Flutter is not just another framework it's a complete SDK that includes everything needed to build visually appealing and fast-performing apps. Unlike traditional Android development, which relies heavily on Java or Kotlin with XML layouts, Flutter uses a widget-based architecture. Widgets are the building blocks of Flutter apps—everything from buttons and text fields to complex layouts and animations is a widget.

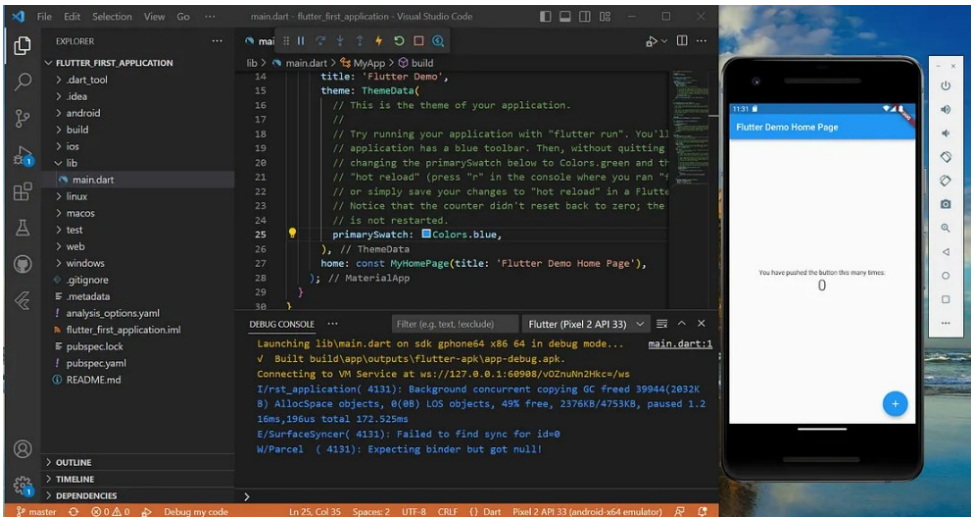


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Fig. 2.7: Setup Flutter framework in vscode

Key Features for Android Development

- Widget Tree and State Management: Apps are built as a tree of widgets. Stateless widgets for static UI, Stateful for dynamic. Providers, Riverpod, or Bloc are popular for managing state in complex apps like reporting tools in "Mukto Pakhi."
- Platform Channels: For Android-specific features not natively in Flutter (e.g., accessing Android's Biometric API), platform channels allow Dart code to communicate with native Java/Kotlin code.
- Plugins and Packages: Flutter's ecosystem is vast. For an app like "Mukto Pakhi," packages like image_picker for photos, geolocator for live locations, url_launcher for emails and calls, and firebase_core for backend integration make development straightforward.

- Performance Optimization: Flutter apps compile to native code, often outperforming hybrid frameworks like React Native. Tools like DevTools help profile CPU, memory, and network usage.
- Testing and Deployment: Unit, widget, and integration tests are built-in. Deployment to Google Play is seamless, with support for flavors (e.g., debug vs. release) and obfuscation for security.

2.4 Dart Programming

Dart is Google's modern, client-optimized programming language that powers Flutter apps. It combines clean syntax, strong safety, and high performance, making it ideal for building fast, reliable Android application that focused on reporting, community collaboration, and real-time alerts.

Core Strengths of Dart in Flutter/Android Context:

- **Object-Oriented & Familiar Syntax**: Dart feels natural if you know Java, Kotlin, or C#. It supports classes, inheritance, mixins, and interfaces perfect for structuring complex logic in apps like Mukto Pakhi.
- **Sound Null Safety**: Prevents null reference crashes at compile time. In Mukto Pakhi, this ensures safer handling of optional fields like photo paths, email recipients, or location data reducing bugs when users submit incomplete reports.
- **Async / Await & Futures / Streams**: Makes asynchronous operations (network calls, file I/O, location updates) feel synchronous and readable. In Mukto Pakhi:
    - Fetching/updating community posts
    - Uploading photos
    - Sending emails
    - Getting live location via GPS all happen without freezing the UI, keeping the app responsive even during slow networks.

- **Isolates (Lightweight Concurrency):** True parallelism without shared memory. Use isolates for heavy tasks (e.g., image processing before upload, parsing large report lists) so the main UI thread stays fluid preventing jank when volunteers browse many cases or upload evidence.

Why Dart Fits Mukto Pakhi Perfectly

In a community-driven Android app handling sensitive real-time data (photos, locations, calls, emails), Dart provides:

- Reliability: null safety + strong typing reduce crashes in critical moments.
- Responsiveness: async/await + isolates keep UI smooth during uploads or background syncing.
- Developer Speed: hot reload + clean syntax accelerate building and fixing volunteer tools.
- Performance: AOT compilation ensures the app feels native on Android devices.

In short, Dart enables Mukto Pakhi to be fast, safe, maintainable, and delightful to use empowering volunteers to protect children effectively without technical friction.

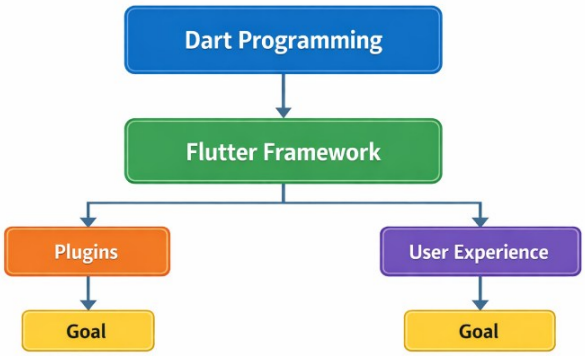

Fig. 2.8: Flow chart diagram for Dart & Flutter overflow

CHAPTER III

Result and Discussion

3.1 Project Overview

Mukto Pakhi is a simple Android application made to help protect children from child labour and abuse in Bangladesh. The project started from the idea that technology can solve real-world problems like child labour, which is a big issue in our country. The main goal of Mukto Pakhi is to make a safer place for children by letting people report problems quickly, build a community, track locations, and spread knowledge about child rights.

The frontend part of Mukto Pakhi is built with Flutter, which makes the app look nice and work fast on Android phones. Users see a simple login screen where they enter email and password, or a create profile page with name, phone, email, password, and address. After login, the home page has buttons for direct report, evidence, call 109 helpline, community contact, map of cases, feedback, and settings. For example, in reporting, users can take photos with the camera plugin or pick images from gallery using image_picker, add location with geolocator, and send alerts with flutter_local_notifications. The map uses flutter_osm_plugin to show reported spots with markers, and url_launcher opens phone calls or emails. Everything is designed to be easy, with Bangla support and smooth on low-end phones.

For the backend, Mukto Pakhi uses Firebase services to keep things secure and real-time. Firebase Authentication handles login and registration, while Cloud Firestore stores user data in five fields like name, phone, email, password, and address. The http plugin connects to outside APIs and provider manages the app's state so screens update quickly. This backend makes sure data is protected with encryption, and only verified users can access or change their profiles from settings.

The working procedure of Mukto Pakhi is straightforward. A user starts by registering or logging in, then goes to the home page and all services will be appears and accessible. This flow turns normal people into helpers, making rescues faster and building trust to fight child labour better.

3.2 User Section

The user section of Mukto Pakhi focuses on how the app looks and feels when running on an actual Android phone. The overall appearance is considered HCD standard design for clean, modern, and very easy to use, even for people who are not very familiar with Android[3]. The app uses a calming teal-green color theme (like the login and profile screens), simple white backgrounds, large readable text, and iconic buttons, all designed to give a friendly and trustworthy look suitable for parents, teachers, and community members in Bangladesh.

3.2.1 Home Page

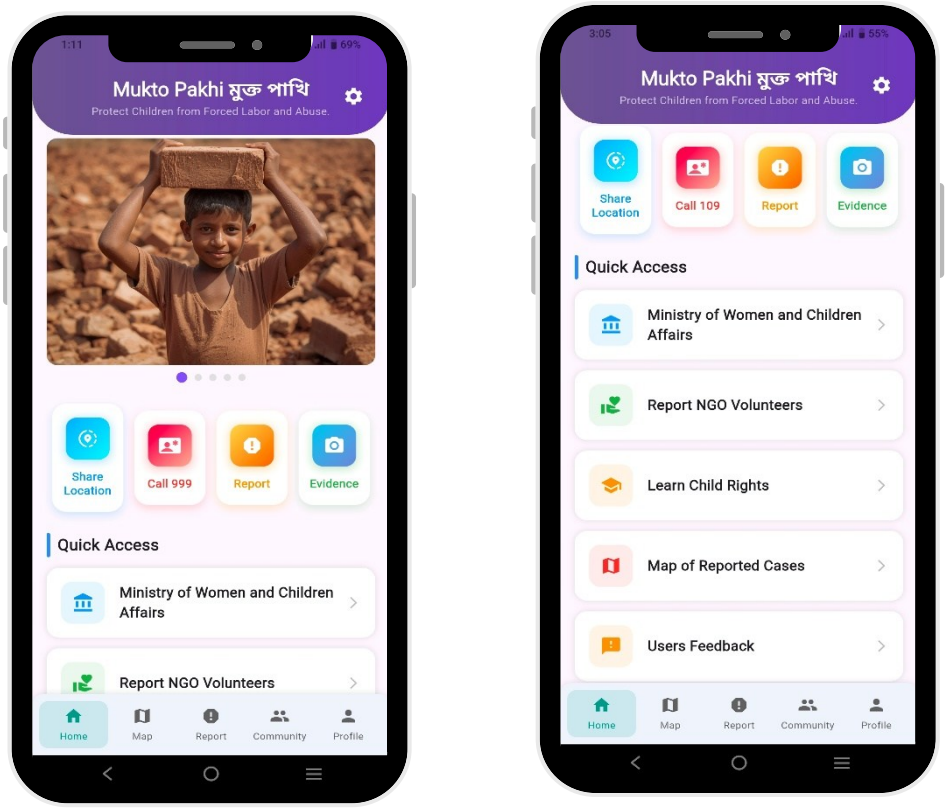

Fig. 3.1: Home page Screens

The Home Page serves as the main dashboard after login. It shows quick access buttons for Direct Report, Evidence, Call 109, Community Contact, Map of Reported Cases, Users Feedback, and Settings. Each button has a clear icon and short label, arranged in a grid or list

that fits nicely on different screen sizes. The design avoids clutter so users can find what they need in one or two taps.Below the header is a powerful full-width image of a young boy carrying a heavy brick on his head, this emotional photo reminds users why the app exists and creates a strong feeling to take action. Under the image is a row of four colourful circular quick-access buttons that stand out clearly:

- Share Location (blue icon): lets users send their current position to trusted contacts or authorities instantly using geolocator.
- Call 109 (red icon): directly opens the phone dialer to call Bangladesh's national child helpline 109 for emergency help.
- Report (orange icon): takes the user to the report page to file a new case of child labour or abuse with evidence.
- Evidence (blue camera icon): opens the section to capture or upload photos/doc. files as proof for reports.

The Quick Access section on the Home Page of Mukto Pakhi gives users fast links to the most important support and learning resources so they can take action or get information without searching through menus. This section is designed with clean white cards, each having a small icon and clear title to show there is more inside. All cards are easy to tap and arranged one below the other for smooth scrolling on any Android phone.

3.2.2 Report Page

The Report page is the most important and powerful section of Mukto Pakhi, where any user can quickly report cases of child labour or abuse to start the rescue process. When a user taps the “Report” button from the home page or bottom navigation, they reach this screen titled “Report to Child Labor”. The page has a clean teal header with a back arrow and a clear title, making it easy to understand and use even on small phone screens.

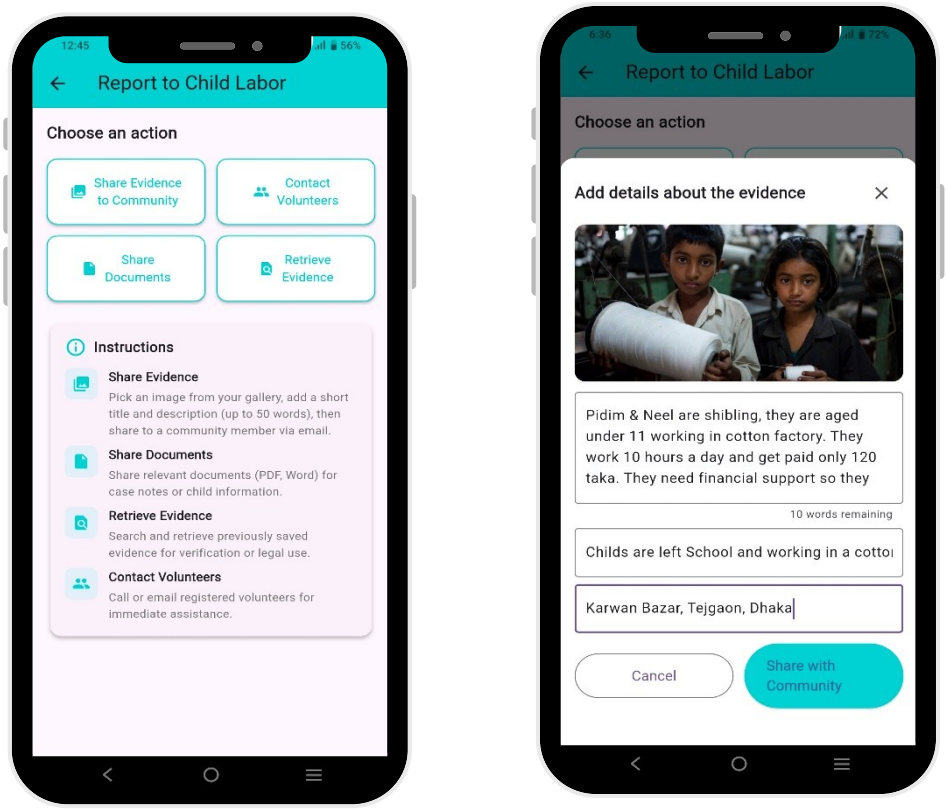

Fig. 3.2: Report page screens

At the top, there is a section called “Choose an action” with four large, colourful buttons that show the main options:

- Share Evidence to Community: allows users to pick a photo from the gallery or take a new one, add a short title and description (up to 50 words), and share it directly with community members or volunteers via email or app notification.

- Contact Volunteers: opens a list of registered local volunteers or NGOs so the user can call or email them immediately for on-ground help.
- Share Documents: lets users upload important files like PDF or Word documents (for example, school leaving proof, medical notes or case file etc) to support the case.
- Retrieve Evidence: helps users search and view previously saved photos, videos, or documents for verification or to add to a new report from database.

This page is made very simple so that parents, teachers, neighbours, or even young users can report fast without confusion. By attaching real photos as strong evidence, adding exact location, and writing a short clear description, the report becomes powerful enough for community members, volunteers, or authorities to verify quickly and take action like rescue or legal steps. The design uses big buttons, clear icons, Bangla support where needed, and works smoothly on low-end Android phones, making it practical for people all over Bangladesh to help stop child labour one report at a time.

3.2.3 Community Build Up

The Community group the heart of Mukto Pakhi, helping to build a strong network of caring people who work together to protect children from forced labour and abuse. When a user taps the “Community” in the bottom navigation bar, they reach the Community screen. This page shows a clean list of registered community members with green circular profile icons, in the example screenshot, members from different areas like Bagerhat, Khulna, and Dhaka are visible, including names, phone and email address. This list makes it easy to see and contact trusted volunteers or helpers nearby.

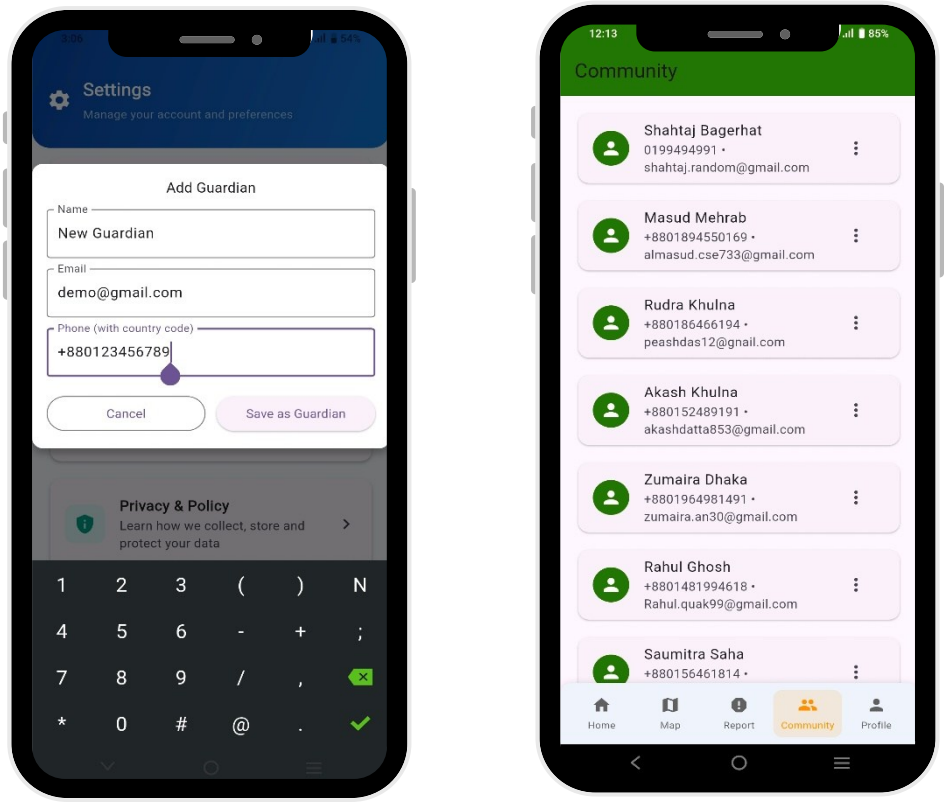

Fig. 3.3: Set Guadian and Community building screens

This community system is very important because child labour cases are often solved faster when more people are watching and helping. For example, when a report is shared, community members get notified instantly and can verify the case, reach the location, or contact authorities together. This feature makes the app not just a reporting tool, but a real community movement for child safety across Bangladesh.

3.2.4 Location Tracking

After a user submits a report in Mukto Pakhi, the app starts a clear and organized process to make sure the case gets proper attention, verification, and quick action. When the report is sent with strong evidence with a short title, detailed description, and exact location its shown on the Map of Reported Cases page using the flutter_osm_plugin. On this map, every report appears as a marker (small pin) at the correct place, along with a card that shows the attached photo, title, description, location name, views count, status and time posted.

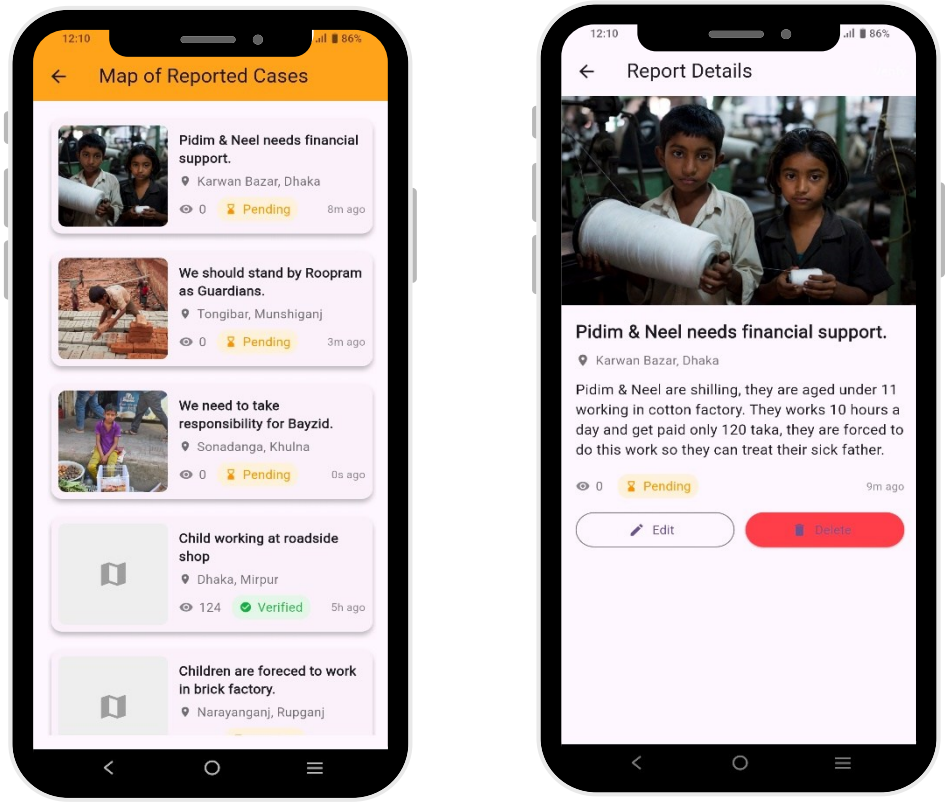

Fig. 3.4: Reported cases and reported case profile

Once the report is live on the map, it is also shared with community members in the same or nearby region. The app uses the location data to notify relevant volunteers, guardians, or local members through push notifications. These community members open the Community page or tap the marker on the map to view the full Case Profile. In the Case Profile, they see the full, evidence photos, location pin on the OSM map, and status. Community members can then analyze the case ,check if the photo and details look real, discuss in the community forum, or

even go to the spot to verify on ground. If everything is correct, they take legal steps forward the case to police, Department of Labour, Ministry of Women and Children Affairs, or NGOs for rescue, counseling, and action against the employer and ensure rehabilitation of the children.

Community members can then analyze the case, check if the photo and details look real, discuss in the community forum, add reviews, or even go to the spot to verify on ground by using OSM map. They can zoom and move the interactive OSM map to see nearby roads, landmarks, or other reported cases, which helps them understand the situation better and plan help faster.

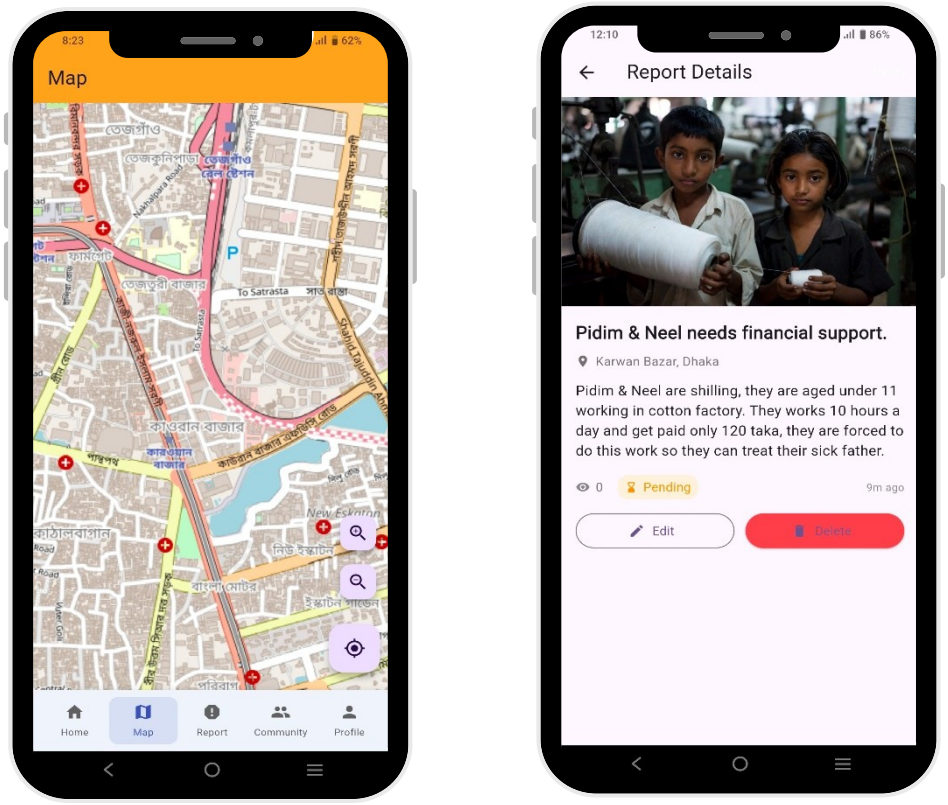

Fig. 3.5: Location tracking through the case profile

This complete flow from submission to community verification, map tracking, active volunteering and administration support makes Mukto Pakhi very effective. It turns one person's report into a group effort with strong evidence, real-time map tracking, and fast official response, helping to rescue children quickly and stop child labour in Bangladesh more successfully.

3.2.5 Additional Tools

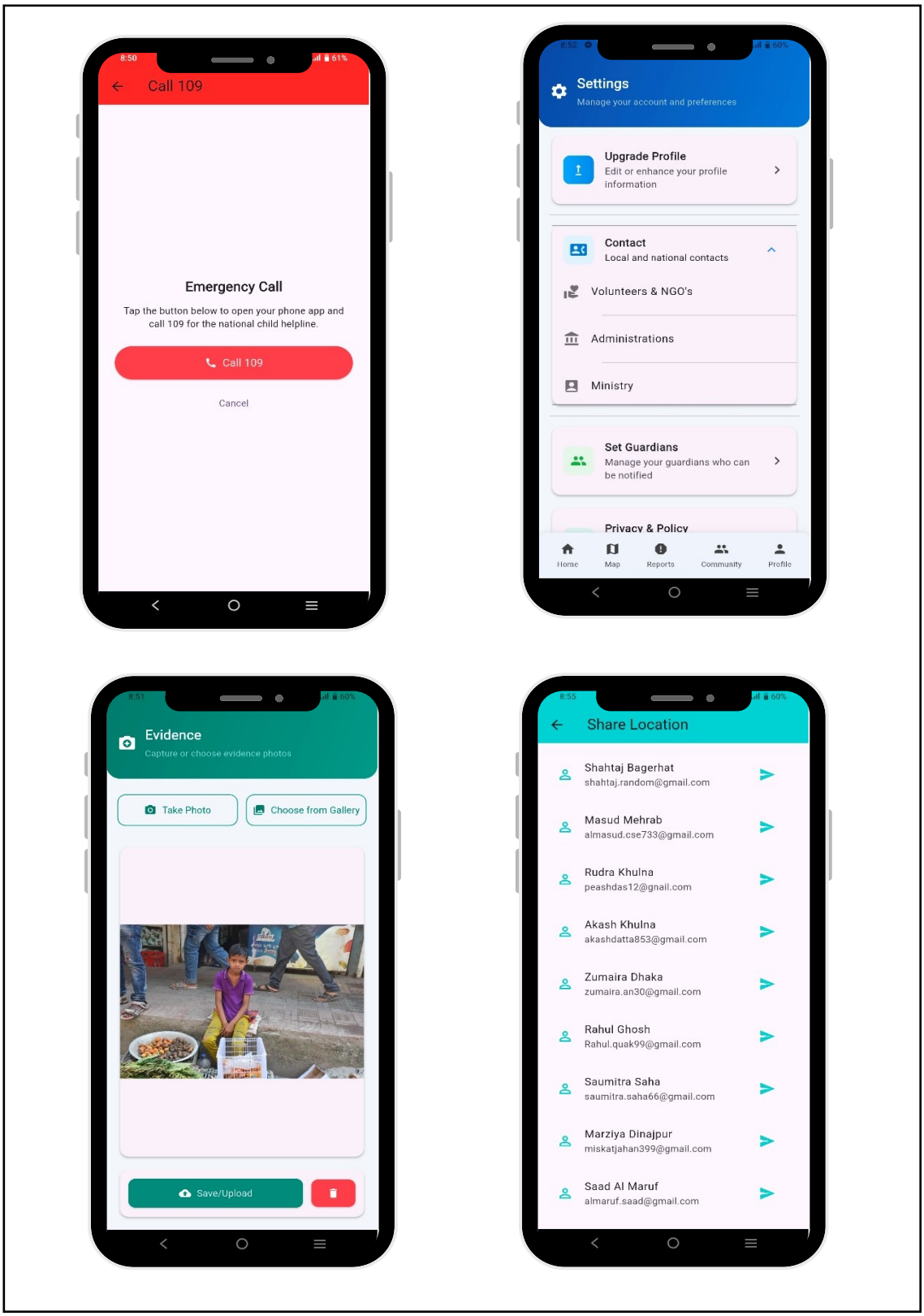

Fig. 3.6: Additional tools and services

Mukto Pakhi includes four additional tools that make the app more efficient, practical, and powerful for protecting children from forced labour and abuse. These features help users act faster, connect with the right people, and keep strong proof for long-term use.

1. **Call 109** – Direct Emergency Helpline From the home page or quick access. Tapping it directly opens the phone dialer and calls Bangladesh's national child helpline (109) for immediate emergency support or complaints.
2. **Contact Administration, Volunteers & NGOs** In the Settings page under "Contact", users find quick links to local and national contacts, registered volunteers, NGOs, administrations, and the Ministry of Women and Children Affairs. These sections let users call, email, or message officials and organizations for suggestions, complaints, opinions, or follow-up on cases.
3. **Share Location with Community** The "Share Location" tool shows a list of community members. Users select one or more trusted people, and the app sends their current location (captured by geolocator) instantly via notification or message. This helps community members reach the incident spot quickly to verify, provide help, or support legal action, making group response much faster and more effective.
4. **Save Evidence** in App Storage In the "Evidence" section, users can take a new photo with the camera or choose from gallery. After previewing the image, they tap "Save/Upload" to store it securely in Firebase Storage or S3 Bucket. Saved evidence can be retrieved later for new reports, added to ongoing cases, or used as proof for complaints to authorities. This keeps important photos safe and ready, strengthening every report.

These four tools work together to give users quick emergency options, direct connections, fast community support, and reliable proof storage turning Mukto Pakhi into a complete, helpful solution for child safety in Bangladesh.

3.2.6 Registration Page

The registration page (titled "Create Profile") in the Mukto Pakhi Flutter app is a simple, clean user onboarding screen designed for new users to quickly set up their account. All the data users enter (name, phone, email, password, address) is securely stored in Firebase Firestore database.

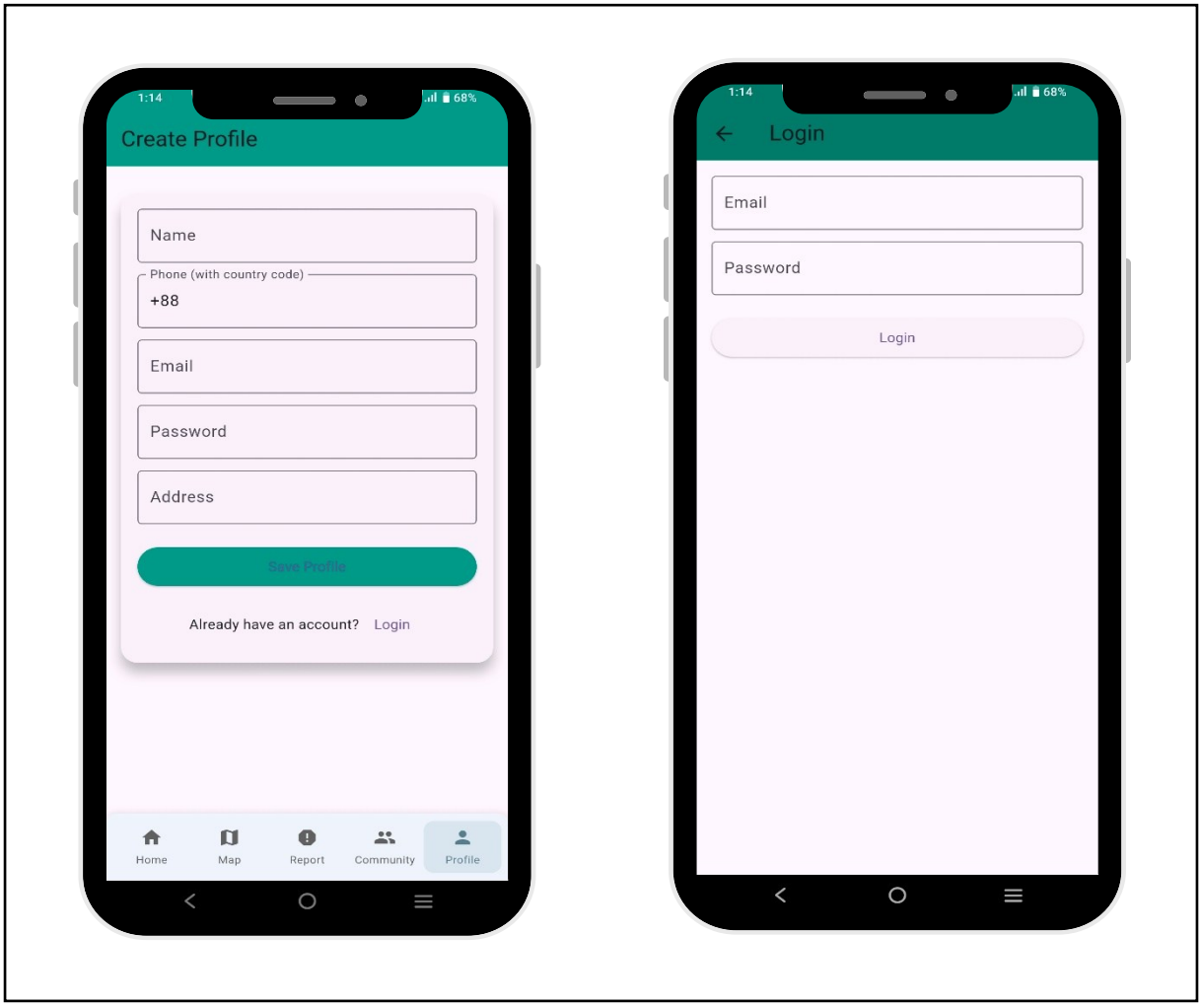

Fig. 3.7: Authentication module screens

Fields are authenticate users information from the above module form are described:

1. **Name**: User's full / display name
2. **Email**: Optional or required for recovery / notifications
3. **Password**: Standard signup password field

4. **Phone (with country code)**: Pre-filled with **+88** (Bangladesh default) that very important field (most likely used for OTP login/verification, Firebase Auth, or as primary identifier in BD apps)
5. **Address**: Probably for location-based features (delivery, community reporting, map-related functions, etc.)

Firebase (Google's backend platform) is widely used in Flutter apps like Mukto Pakhi for:

- Secure user authentication (email/password or phone-based)
- Encrypted storage of profile data in the cloud
- Automatic HTTPS protection
- Firebase Security Rules to control who can read/write data (only the authenticated user can access their own profile)

This ensures your information stays private, protected from unauthorized access, and is handled according to modern security standards. After saving, you'll likely be logged in automatically and redirected to the Profile or Home screen.

3.2.7 Login Page

A login page is the entry gate for users who already have an account. It asks for credentials usually like email and password to verify your identity and let you into the app as "you" (instead of as a guest). The Login page in the Mukto Pakhi app is a straightforward screen that allows existing users to sign in securely with only email and password and access their profile, reports, community features, or other personalized parts of the app.

The login form require only two simple input fields:

- Email (text field for entering your registered email)
- Password (secure text field, usually masked with dots/asterisks)

The Login page in the Mukto Pakhi app is a straightforward screen that allows existing users to sign in securely with only email and password and access their profile, reports, community features, or other personalized parts of the app.

CHAPTER IV

Conclusion

After developing the Mukto Pakhi project during my industrial attachment, I have reached several meaningful conclusions. The project has provided valuable experience in full-stack Flutter app development with Dart programming and Firestore database authentication expertise also UI/UX design concepts. Through this hands-on work, I learned how to build a user-friendly mobile application that connects communities for social good and solve real-word problem.

The main goals of the app include raising community awareness, enabling early detection and reporting of child abuse or forced labor, supporting quick rescue operations, and linking rescued children to rehabilitation services such as education, counseling, medical care, and family reunification or guardianship. By using Firebase for secure data storage and authentication, the app ensures user privacy and reliable information flow, which is very important for sensitive reports.

The positive impact of Mukto Pakhi can be huge if widely used. It empowers ordinary people like students, parents, teachers, or neighbors to become part of the solution, turning passive observation into active protection. Even one successful report can save a child from lifelong harm, restore their right to education, and break the cycle of poverty and exploitation. In the long run, it contributes to national efforts like the National Plan of Action to Eliminate Child Labour (2021–2025) and global targets (SDG 8.7) to end child labor by promoting faster rescues, better rehabilitation, and stronger community guardianship systems[4].

Overall, this project has taught me that technology, when combined with social purpose, can create real change. Mukto Pakhi is not just an app, it is a small but important step toward a Bangladesh where every child is safe, free from abuse and forced work, and able to grow with dignity and hope. I believe more such initiatives, supported by government, NGOs, and youth like us, will help build a brighter future for our vulnerable children.

CHAPTER V

Future Work

In the upcoming days, the Mukto Pakhi app will see several advanced features and improvements to make it more powerful, secure, and impactful in rescuing children from forced labor and abuse while supporting their rehabilitation and guardianship. These future works will help expand the app's reach and effectiveness in Bangladesh

In the future, I plan to develop the following features for Mukto Pakhi:

1. Donation System: A secure donation feature will be added, so users, individuals, organizations, or companies can easily contribute money for impoverished children's rehabilitation, education, counseling, medical care, and volunteering rescue operations. It will support various payment methods popular in Bangladesh (like bKash, Nagad, Rocket, bank cards, and mobile banking) through trusted gateways. This will create a sustainable funding source for NGOs and partners, encouraging community involvement and making real help faster for rescued children.

2. Advance Documentation: Users will be able to upload or record videos and live streaming of incidents directly in the app. These evidences will be securely saved in our cloud storage (Firebase Storage or similar secure solution) with proper encryption and access controls. This will strengthen case reports, help authorities verify facts quickly during rescues, and provide stronger proof for legal actions or NGO interventions, reducing false reports and improving response speed.

3. AI Chatbot: An intelligent AI-powered chatbot (built with tools like Gemini, Grok, or custom models integrated via Flutter) will be introduced. It will handle user queries about child rights, guide on how to report, analyze case details for patterns, verify or review incoming reports automatically (e.g., check for duplicates or urgency), and search old reports/evidence from the Firestore database. This will make the app smarter, provide 24/7 support, reduce workload on human moderators, and help users or officials get quick insights while protecting sensitive data.

4. Enhance Security: As user numbers and evidence storage grow large, security will be a top priority. Future updates will include stronger measures like advanced encryption for all data (at rest and in transit), better Firebase Security Rules, two-factor authentication (2FA), regular security audits, role-based access (e.g., only verified NGOs see full details), anonymization where possible, and protection against common threats like data breaches or misuse.

5. iOS & Web Application: Soon, we will develop and launch versions for iOS (using the same Flutter codebase for smooth performance) and a progressive web app (PWA) that works on any browser (desktop, laptop, or mobile). This will allow more people including those without Android phones, teachers, parents, or officials using computers to access reporting, donation, and awareness features easily.

These upcoming advancements will turn Mukto Pakhi into a more complete, technology-driven platform for child protection. By combining community reporting, secure tools, AI support, funding, and wider accessibility, the app will contribute more strongly to national goals like ending child labor and protecting every child's rights in Bangladesh.

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[3] Human-Centered Design in Mobile Application Development: International Journal of Mobile Human Computer Interaction(October 2014)- DOI:10.4018/jmhci.2012100101, Authors: Eyal Eshet at Åbo Akademi University

[4] ILO Contributions to achieve SDG Target 8.7: Ending Child Labour – Implementation of SDG 8.7 Simela Papatheophilou, Bernhard Tröster, Vienna, September 2025.